

Introduction to Linguistics

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Outline

- 1 Introduction
- 2 Phonetics and Phonology
- 3 Morphology
- 4 Syntax
- 5 Semantics
- 6 Pragmatics and Discourse

Introduction

What is Linguistics?

Linguistics is the scientific study of language. It seeks to answer the following questions:

- What exactly do we know when we know a language?
- How is this knowledge acquired?
- How is such knowledge used?

It also investigates:

- Which characteristics are unique to a particular language?
- Which characteristics are universal and found across all languages?
- How do individual languages differ?
- How and why do languages change, emerge, or die?

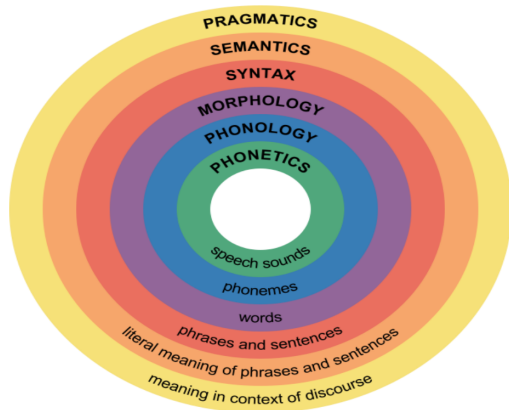
Interesting Facts About Language

- Language is a unique ability of humans.
- Humans can express and understand complex thoughts within seconds.
- Children acquire language naturally within the first few years of life.
- The number of possible sentences in a language is effectively infinite.
- We can distinguish between grammatical and ungrammatical sentences.
- We can understand truncated sentences that omit words, e.g., *Stop it!*
- We can recognize ambiguity in sentences, e.g.,
John saw the girl with binoculars.
- We can identify sentences that paraphrase each other while conveying the same meaning.
- Language allows creativity, productivity, and flexibility in communication.

Formal Language vs. Natural Language

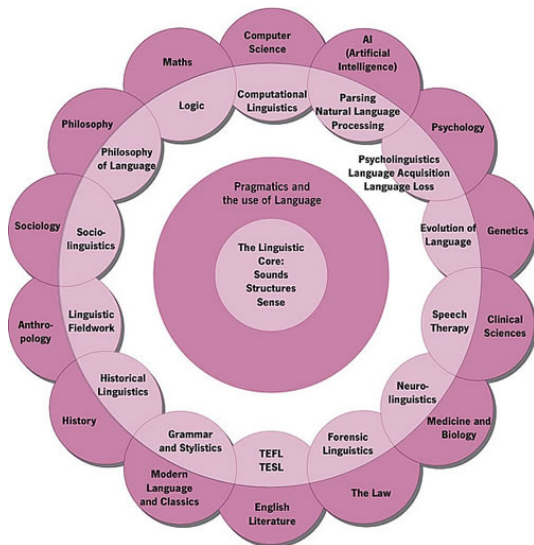
Feature	Formal	Natural
Origin	Invented	Evolved
Grammar	Defines language	Emerges from language
Learning	Rule-based	Naturally acquired
Meaning	Unambiguous	Ambiguous
Context	Independent	Dependent
Interpretation	Machines	Humans
Knowledge	No world knowledge	Uses world knowledge
Change	Stable	Dynamic
Examples	Python, Logic	Spoken Language, ASL

Levels of Linguistics Analysis



Major Levels of Linguistic Structure

Applied Linguistics



Levels of Linguistics Analysis

- Phonology (Study of sounds)

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Levels of Linguistics Analysis

- Phonology (Study of sounds)
- Morphology (Study of words)
- Syntax (Study of sentence structure)
- Semantics (Study of meaning)
- Pragmatics (Study of Language Use)

Phonetics and Phonology

Phonetics

Phonetics is the study of the physical sounds of human speech. It examines how speech sounds are produced, transmitted, and perceived.

We can study the sounds of language from the perspective of:

- ➊ **Articulatory Phonetics:** Studies how speech sounds are produced by the movement of speech organs (e.g., tongue, lips, vocal cords).
- ➋ **Acoustic Phonetics:** Analyzes the physical properties of speech sounds as sound waves, such as frequency, amplitude, and duration.
- ➌ **Auditory Phonetics:** Investigates how speech sounds are perceived by the ear, auditory nerve, and brain.

Distinguishing Speech Sounds

Speech sounds are distinguished by three major features:

① Voicing

- ▶ Whether the vocal folds vibrate during the production of a sound.
- ▶ Example: /b/ (voiced) vs. /p/ (voiceless)

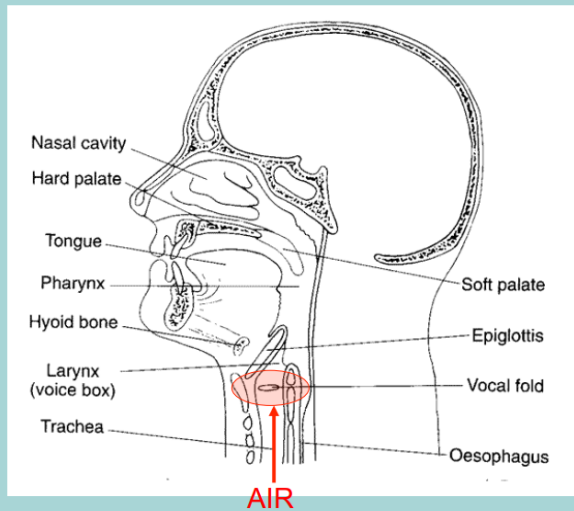
② Place of Articulation

- ▶ Where the airflow is constricted in the vocal tract.
- ▶ Example: Bilabial, Dental, Alveolar, Velar

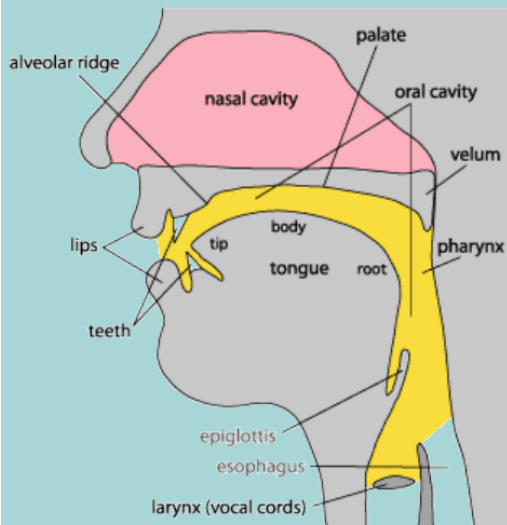
③ Manner of Articulation

- ▶ How the airflow is modified during speech production.
- ▶ Example: Stop, Fricative, Nasal, Approximant

The vocal tract



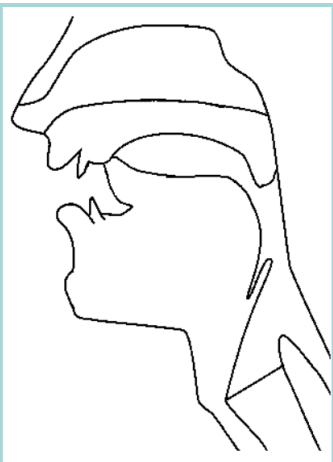
Points of articulation



Manner of articulation

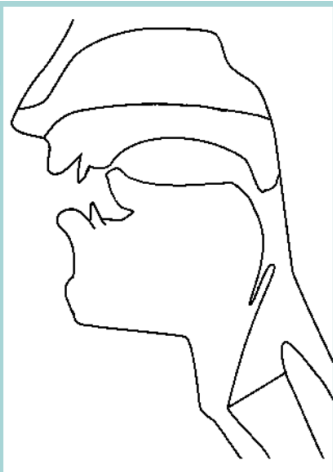
Alveolars

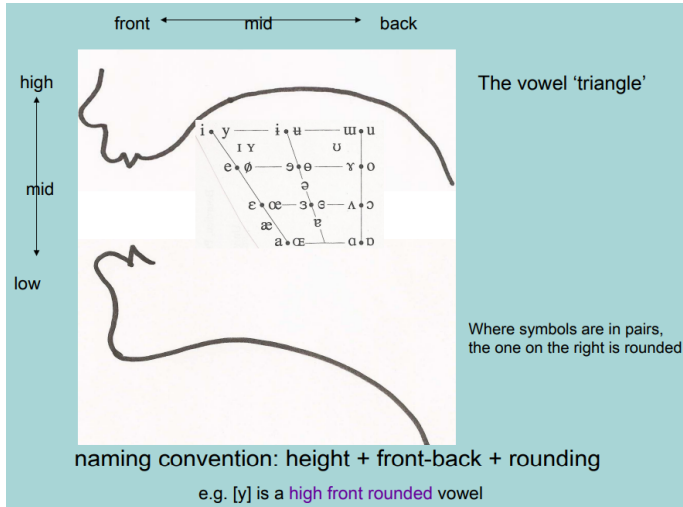
stop	t
nasal	n
trill	r
flap	ɾ
fricative	s
lat. fricative	ɬ
affricate	tʃ
approximant	ɹ
lat. approx.	ɻ



Manner of articulation

stop	t
nasal	n
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International Phonetics Alphabet

THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

CONSONANTS (PULMONIC)

© 2020 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

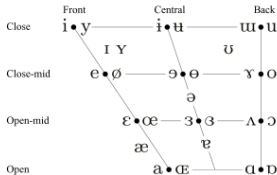
Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

CONSONANTS (NON-PULMONIC)

Clicks	Voiced implosives	Ejectives
◌ǀ Bilabial	◌ɓ Bilabial	◌' Examples:
◌ǃ Dental	◌ɗ Dental/alveolar	◌p' Bilabial
◌ǂ (Post)alveolar	◌ɟ Palatal	◌t' Dental/alveolar
◌ǁ Palatoalveolar	◌ɡ Velar	◌k' Velar
◌ǁ Alveolar lateral	◌ɠ Uvular	◌s' Alveolar fricative

◌ǀ ◌ǃ ◌ǂ ◌ǁ ◌ǁ

VOWELS



Phonology

Phonology is the study of how sounds function within a particular language or languages. It focuses on the abstract, cognitive aspects of sounds and how they are organized in the mind.

- Concerned with the abstract, mental representation of sounds in a specific language.
- Studies the rules and patterns by which sounds are combined and used to convey meaning.

Phonology

- **Phonemes:** The smallest units of sound that can distinguish meaning in a language (e.g., /p/ and /b/ in "pat" vs. "bat").
- **Allophones:** Variations of phonemes that do not change the meaning of words and occur in specific phonetic contexts (e.g., the aspirated [p] in **pat** and the unaspirated [p] in **spat**).

Phonemes & Allophones

Phoneme / slash notation /

The smallest sound unit that **changes meaning**.

Minimal Pairs- English:

pat / *bat* /p/ ≠ /b/

sip / *zip* /s/ ≠ /z/

cat / *cut* /æ/ ≠ /ʌ/

Swap the sound → meaning changes
∴ they are different phonemes.

Allophone [bracket notation]

A variant of a phoneme.
Never changes meaning.

English /p/ same phoneme, 3 sounds:

pin [p^h] â aspirated

spin [p] â unaspirated

tip [p̚] â unreleased

Same phoneme /p/.

Phoneme = the idea of a sound. Allophone = how it is actually pronounced.

Phonetics vs Phonology

Phonetics

- Physical aspects Speech sounds. /p/
- Do not take meaning into account
- Base for phonological analysis
- Analyses the production of all speech sounds regardless of language [Phone]
- Phonetic Transcription []

Phonology

- Abstract/cognitive
- Takes meaning into account
- Base for morphological and further analysis
- Analyses sound pattern of particular language /Phoneme/
- Phonemic transcription //

Morphology

Morphology

Two dominant views of Morphology:-

Morphology is the study of

- the mental system involved in word formation
- words, their internal structure, and their formal relationships

Its etymology is Greek:

- *morph* - means 'shape, form'
morphology is the study of form or forms.
- The word '*morphologie*' was first used by August Schleicher in 1859.
- The earliest and the first morphological analysis of a human language: Ashtadhyayi by Panini.

Three kinds of Signs

1. Icon

The icon is the sign which physically resembles what it 'stands for'.

Form:



Meaning: tree

2. Index

Index has a correlation in space and time with its meaning.

Form:



Iconic Meaning: Skull and Bones

Indexical Meaning: Danger

Three kinds of Signs

3. Symbols

- Form and Meaning are related only arbitrarily.
- The relation is convention.
- Words are symbols.

Form: w-a-t-e-r

Meaning: water

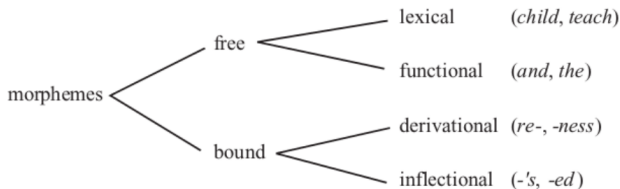
Concepts of Morphology

- Native speakers create new words from the existing ones or borrow from other languages as and when necessary.
- The discovery of these mechanisms and the intuitive knowledge underlying this creativity is what is usually known as morphology.
- Native speaker's ability to derive or relate the words in terms of their form and meaning.
- Alternatively native speakers ability to reject *cat-en, *cat-z, *cat-iz, (for cats), walk *walken; drive *drived; read *readed, *readen; active *activement, *activance, and *activant as illformed is because of the knowledge of morphology.

Building Blocks of Morphology

Morpheme: the smallest meaningful linguistic unit. Some morphemes are identical with words, but many morphemes are smaller than words.

Morpheme $\leq$ Word



Free morpheme: a morpheme that can stand alone, that is a complete word; an independent morpheme. eg. walk, book, but, of and etc.,

- **Lexical morpheme:** a morpheme that denotes the content words. They receive inflection.
 - ① **Open class** categories such as Nouns, Verbs and Adjectives.
 - ② **Closed class** categories such as Pronouns, Number words and Nouns of space and time.
- **Functional morpheme:** a morpheme that denotes the functional words.
 - ▶ They do NOT receive inflection and are indeclinables or avyayas.
 - ▶ Categories such as Prepositions/Postpositions, Conjunctions, Interjections, Adverbs, Demonstratives, Intensifiers, Quotatives etc,

Bound morpheme: a morpheme that cannot stand alone, but must be attached to something else. Ex. -ed, -s, -er

- **Inflectional Morpheme:** creates new forms of the same word with the addition of grammatical properties; the basic meaning (and the category) of the word is the same.

Inflection in English:

- a. with nouns: book, books, book's, books'
- b. with verbs: ride, rides, rode , ridden, riding
- c. with adjectives: old, older, oldest
- d. with numbers: seven, seventh

- **Derivational morpheme:** creates a new word with a different meaning that may belong to a different or to the same grammatical category.

RE + WRITE = rewrite 'write again', verb

WRITE + ER = writer 'one who writes', noun

Derivation in English:

- a. Verb to Noun: kill = killer
- b. Noun to Verb: glory = glorify
- c. Adjective to Noun: dark = darkness
- d. Noun to Adjective: person = personal
- e. Adjective to Verb: modern = modernize
- f. Verb to Adjective: walk = walkable
- g. Adjective to adverb: great = greatly

Affixes

Affix

- An affix is a **bound morpheme** attached to a stem or root to form a word.

Types of Affixes

Derivational Affixes

- Change meaning or word class
- boy + *-ish* → boyish
- care + *-less* → careless

Inflectional Affixes

- Express grammatical information
- book + *-s* → books
- work + *-ed* → worked

Concatenative Morphology

Concatenative Processes

- Affixes are added linearly before or after a root/stem.

Type	Definition	Example
Prefix	Added before a root/stem	un + do → undo
Suffix	Added after a root/stem	look + ing → looking

Schemas

- Prefix + Root/Stem
- Root/Stem + Suffix

Non-Concatenative Morphology

Type	Definition	Example
Infix	Inserted within a stem	bili 'buy' → bum ili 'bought'
Circumfix	Two-part affix surrounding a stem	berg → ge-berg-te

Examples of Circumfixes in Dutch

- berg 'mountain' → gebergte 'mountains'
- vogel 'bird' → gevogelte 'birds'
- raam 'frame' → geraamte 'frames'

Allomorph

The physical variation of a morpheme is called as allomorphs.

A morpheme may display allomorphy, i.e. have more than one form.

The allomorphs are physically different forms, but they indicate same meaning of a morpheme.

Types of Allomorph:

- Phonologically Conditioned Allomorph
- Lexically Conditioned Allomorph
- Suppletive

Phonologically Conditioned Allomorph

The variants in the pronunciation of the plural are phonologically conditioned allomorphs, because the choice depends only on the phonological characteristics of the element to which it attaches.

The English plural morpheme has three allomorphs:

- /s/ bits, tips, tacks
- /z/ dogs, slabs
- /ɪz/ ladies, bodies

Here, [s], [z], [ɪz] are the allomorphs of the morpheme, let's say /z/.

Lexically Conditioned Allomorph

When the choice of allomorphs are unpredictable from the knowledge of the language's Morphology and Phonology, they are lexically conditioned allomorphs.

Since there is no pattern, there is no rule: the information is just in the lexicon.

English:

two oxen	*two oxes	*two ox
two deer	*two deers	*two deeren
man	men	
child	children	
sheep	sheep	

Suppletive Allomorph

Suppletion is an extreme form of allomorph in which two completely different roots realize the same morpheme.

Examples:

go went

be is - was - were - am

good better - best

bad worse - worst

Morphological Typology

Morphological typology is the basis for the broad classification of Languages of the world into four major Morphological types:

Isolating Languages (Analytic):

- Each word tends to consist of a single, independent morpheme
- There are no bound forms i.e. affixes
- Grammatical markers are realized as unattached (free) morphemes
- morpheme = word
- Grammatical changes are indicated by word order

Analytical languages are most common in Southeast Asia (Chinese, Vietnamese), but some such languages are also found among the Austronesian languages (Fijian, Tongan) and some Niger-Congo languages (Gbe, Yoruba).

Morphological Typology

Example from Vietnamese:

no	se	khong	doc	sach
he	FUT	NEG	read	book

'he will not read book'

Example from Chinese:

Ta	ba	shu	mai	le
He	NOM	book	buy	Asp

'He bought the book.'

Morphological Typology

Agglutinative Languages (Synthetic):

- All bound forms are affixes
- They are added to a stem like beads on a string
- Every affix represents different morphological feature
- Each morpheme represents only one grammatical meaning
- morpheme < word
- Word order is slightly less important than it was in analytic languages

e.g. Dravidian, Turkish, Finnish, Hungarian etc.

Morphological Typology

Example from Telugu (Dravidian):

illu	'house'	rA	'to come'
iMti-ni	'house (object)'	vacc-A-nu	'I came'
iMti-ki	'to the house'	vacc-A-mu	'we came'
iMti-lo	'in the house'	vacc-A-vu	'you (sg.) came'
iMti-wo	'with the house'	vacc-A-ru	'you (pl.) came'
...		vacc-A-du	'he came'

Morphological Typology

Inflectional Languages (Fusional):

- Distinct features are merged into a single bound form (portmanteau morph)
- Morpheme boundaries are difficult to identify
- Every suffix has several grammatical functions

The classical Indo-European languages like Sanskrit, Greek, Latin etc. are examples of flexional languages where the inflectional morphemes are said to be “fused” together.

Morphological Typology

Incorporating Languages (Polysynthetic):

- All bound forms are affixes
- Inflections are incorporated into the word
- Ability to form words that are equivalent to whole sentences in other languages
- Morphologically extremely complex
- Generally, morphology is more important than context and syntax

e.g. Icelandic/Aleutian

Inuktitut: *tavvakiquitiqarpiit* → “Do you have any tobacco for sale?”

Yup'ik (Alaska):

angya-	li-	ciq-	sugnar-	quq-	llu
boat	make	FUT	PROB	3sg.NOM	also

‘Also, he probably will make a boat’

Syntax

Introduction to Syntax

- Syntax refers to the set of rules, principles, and processes that govern the structure of sentences in a language.
- These rules determine how words and phrases are combined to form well-structured sentences.
- **Well-formedness vs Ill-formedness:**
 - ▶ A sentence is well-formed if it follows the syntactic rules of the language.
 - ▶ **Example:**
 - ★ *I am happy.* (Well-formed)
 - ★ **I saw girl the tall.* (Ill-formed â incorrect word order)
 - ▶ Syntax helps identify these grammatical errors by enforcing the structure.

Key Concepts in Syntax

- **Syntactic Categories:** These are categories of words that have similar syntactic behavior.
 - ▶ **Lexical Categories:** These include content words such as:
 - ★ Nouns: Refers to people, objects, or ideas (e.g., *boy, school, idea*).
 - ★ Verbs: Describes actions or states (e.g., *run, think, is*).
 - ★ Adjectives: Provides descriptive qualities of nouns (e.g., *big, red, fast*).
 - ★ Adverbs: Modifies verbs, adjectives, or other adverbs (e.g., *quickly, very, always*).

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 - ▶ **Functional Categories:** These serve grammatical purposes and include:
 - ★ Determiners: Specifies a noun (e.g., *the, a, an*).
 - ★ Prepositions: Shows relationships between nouns (e.g., *in, on, under*).
 - ★ Conjunctions: Connects words, phrases, or clauses (e.g., *and, but, because*).
- **Example:**
 - ▶ *The cat sat on the mat.*
 - ★ *The* (Determiner), *cat* (Noun), *sat* (Verb), *on* (Preposition), *the mat* (Noun Phrase).

Phrasal Categories

- **Noun Phrase (NP):**
 - ▶ A group of words that functions as a noun in a sentence.

Phrasal Categories

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- ▶ A group of words that functions as a noun in a sentence.
- ▶ **Examples:**
 - ★ *John* (Simple NP)
 - ★ *The little boy* (NP with Determiner and Adjective)
 - ★ *The cat on the mat* (NP with Prepositional Phrase)
- ▶ **Phrase structure rule:**

$$NP \rightarrow (Det) (Adj) * N (PP) \quad (1)$$

Phrasal Categories

- **Verb Phrase (VP):**

- ▶ A phrase that includes a verb and its complements or modifiers.

Phrasal Categories

- **Verb Phrase (VP):**

- ▶ A phrase that includes a verb and its complements or modifiers.
- ▶ **Examples:**
 - ★ *She runs* (Simple VP)
 - ★ *John sang a song* (VP with Object)
 - ★ *John sang a song in the shower* (VP with Prepositional Phrase)
- ▶ **Phrase structure rule:**

$$VP \rightarrow V (NP) * (PP) (Adv)* \quad (2)$$

Phrasal Categories

- **Prepositional Phrase (PP):**

- ▶ A phrase that begins with a preposition and is followed by a noun phrase.
- ▶ Examples:
 - ★ *in the house*
 - ★ *with a friend*
- ▶ **Phrase structure rule:**

$$PP \rightarrow P \ NP \quad (3)$$

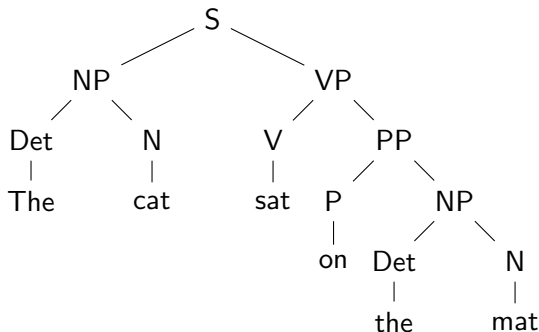
Phrase Structure Grammar (PSG)

- **Phrase Structure Grammar (PSG)** is a type of generative grammar that describes the syntactic structure of sentences using phrase structure rules.

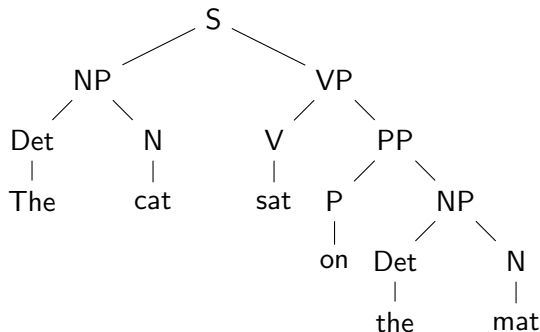
Phrase Structure Grammar (PSG)

- **Phrase Structure Grammar (PSG)** is a type of generative grammar that describes the syntactic structure of sentences using phrase structure rules.
- It breaks down a sentence into its constituent parts, or phrases, which are categorized into syntactic types (e.g., NP, VP, PP).
- **Example Sentence:** *The cat sat on the mat.*

Phrase Structure Grammar (PSG)



Phrase Structure Grammar (PSG)



- **Phrase Structure Rule:**

$$S \rightarrow NP \ VP$$

$$NP \rightarrow Det \ N$$

$$VP \rightarrow V \ PP$$

$$PP \rightarrow P \ NP$$

Sentence Structure in Syntax

- A sentence must contain both a Noun Phrase (NP) and a Verb Phrase (VP).
- The most basic sentence structure can be described using the following phrase structure rule:
 - ▶ $S \rightarrow NP VP$
 - ▶ **Example:**
 - ★ The dog barked.
 - ★ NP = The dog
 - ★ VP = barked
- A sentence is ill-formed if it does not follow this basic structure.
- **Example of Ill-formed Sentences:**
 - ▶ *The dog. (Missing VP)
 - ▶ *Barked. (Missing NP)

Types of Clauses and Sentences

- **Types of Clauses:**

- ▶ **Independent Clause:** A clause that can stand alone as a sentence (e.g., I went to the store).
- ▶ **Dependent Clause:** A clause that cannot stand alone and depends on the main clause (e.g., If I go out).

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- **Types of Sentences:**

- ▶ Simple Sentence: Contains one independent clause (e.g., I like pizza).
- ▶ Compound Sentence: Contains two or more independent clauses (e.g., I like pizza and he likes pasta).
- ▶ Complex Sentence: Contains one independent clause and at least one dependent clause (e.g., laughed when he fell).
- ▶ Compound-Complex Sentence: Contains at least two independent clauses and one or more dependent clauses (e.g., I laughed when he fell, but he was fine).

Complexities in Syntax

- **Ambiguities:**

- ▶ **Structural Ambiguity:** Occurs when a sentence can be parsed in more than one way (e.g., I saw a man with a telescope).
- ▶ **Coordination Ambiguity:** When the scope of conjunctions like "and" is unclear (e.g., old men and women).

- **Garden-Path Sentences:** Sentences that lead to an incorrect initial interpretation (e.g., *The old man the boat*).

- **Recursiveness:** A structure that repeats within itself (e.g., *This is the cat that ate the rat that ate the cheese*).

- **Ellipsis:** Omission of words understood in context (e.g., *I did homework; he didn't*).

Semantics

Semantics

Semantics - Study of meaning

Semantics is that level of linguistic analysis where meaning is analyzed in a language.

- The study of semantics looks at how meaning works in language, and it often uses native speaker intuitions about the meaning of words and phrases in a discourse.

Speaker's Meaning and Sentence meaning

- **SPEAKER MEANING** is what a speaker means (i.e. intends to convey) when he uses a piece of language.
- **SENTENCE MEANING (or WORD MEANING)** is what a sentence (or word) means, i.e. what it counts as the equivalent of in the language concerned.

E.g. A news headline -

“Government to drop ‘White Bomb’ in the next budget.”

In this headline speaker meaning is that milk consumers will face a brunt of change in dairy tax structure.

Meaning Representation

- Meaning Representation involves representing the meaning of natural language in a form that machines can process and manipulate.
- It is crucial for enabling deeper understanding and inference in tasks like machine translation, question answering, and summarization.
- Two major approaches in meaning representation:
 - ▶ Logical Semantics
 - ▶ Semantic Role Labeling (SRL)

Challenges in Meaning Representation

- Natural language is ambiguous, context-dependent, and flexible.
- Meaning must be captured in a formalized, precise structure to enable:
 - ▶ Inference
 - ▶ Reasoning
 - ▶ Question answering
- Complexities include handling negation, quantifiers, modality, and disambiguation.

Logical Semantics Overview

- Logical Semantics is the study of meaning using formal logic.
- It provides a precise, unambiguous structure for representing sentence meaning.
- It helps us answer questions like:
 - ▶ What are the truth conditions of a sentence?
 - ▶ How do we represent meaning in a form that computers can process?
- The most common formalism is First-Order Logic (FOL).

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- The most common formalism is First-Order Logic (FOL).

Components of First-Order Logic (FOL)

- Constants: Represent specific entities (e.g., 'John').
- Predicates: Express properties or relations between entities (e.g., Loves(John, Mary)).
- Variables: Represent arbitrary entities (e.g., $\forall x$, $\exists y$).
- Quantifiers: Express quantities (e.g., $\forall x$, $\exists y$).
- Connectives: Logical operators (e.g., AND, OR, NOT).

Example of First-Order Logic (FOL)

Sentence: *John loves Mary.*

- Logical Form: 'Loves(John, Mary)'
- Predicate: 'Loves'
- Arguments: 'John' and 'Mary'
- Interpretation: This formal representation can be used for reasoning, querying, and making inferences.

What is Semantic Role Labeling (SRL)?

- Semantic Role Labeling (SRL) is a process that identifies the roles of words in a sentence (who did what to whom).
- It captures the relationships between the predicate (verb) and its arguments (subject, object, etc.).
- SRL helps in understanding the meaning of sentences in tasks like machine translation, summarization, and question answering.

Key Concepts in SRL

- Predicate: The main verb or action in the sentence.
- Arguments: The participants involved in the action.
- Semantic Roles:
 - ▶ Agent: The doer of the action.
 - ▶ Patient: The entity affected by the action.
 - ▶ Instrument: The tool used to perform the action.
 - ▶ Location: Where the action occurs.

Thematic Role	Definition	Example
Agent	The volitional causer of an event	<i>The waiter</i> spilled the soup.
Experiencer	The experiencer of an event	<i>John</i> has a headache.
Force	The non-volitional causer of the event	<i>The wind</i> blows debris from the mall into our yards.
Theme	The participant most directly affected by an event	Only after Benjamin Franklin broke <i>the ice</i> ...
Result	The end product of an event	The city built <i>a regulation-size baseball diamond</i> ...
Content	The proposition or content of a propositional event	Mona asked <i>âYou met Mary Ann at a supermarket?â</i>
Instrument	An instrument used in an event	He poached catfish, stunning <i>them with a shocking device</i> ...
Beneficiary	The beneficiary of an event	Whenever Ann Callahan makes hotel reservations <i>for her boss</i> ...
Source	The origin of the object of a transfer event	I flew in <i>from Boston</i> .
Goal	The destination of an object of a transfer event	I drove <i>to Portland</i> .

Table: Thematic roles with their definitions. ¹

¹Speech and Language Processing. Daniel Jurafsky & James H. Martin.

Example of Semantic Role Labeling (SRL)

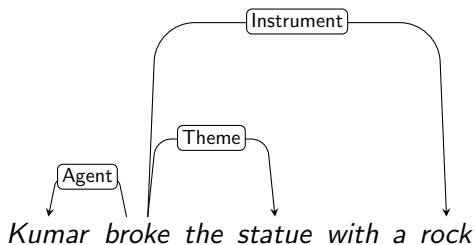


Figure: 1

Example of Semantic Role Labeling (SRL)

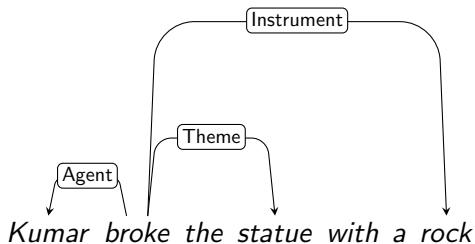
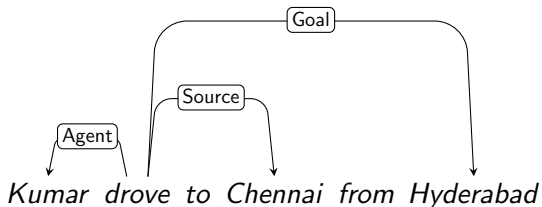


Figure: 1



Example of Semantic Role Labeling (SRL)

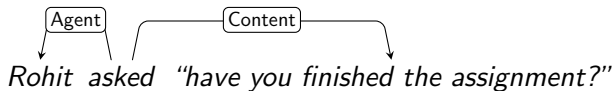


Figure: 3

Example of Semantic Role Labeling (SRL)

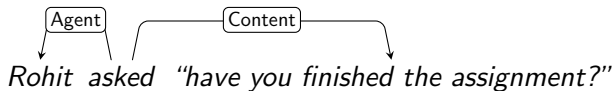


Figure: 3

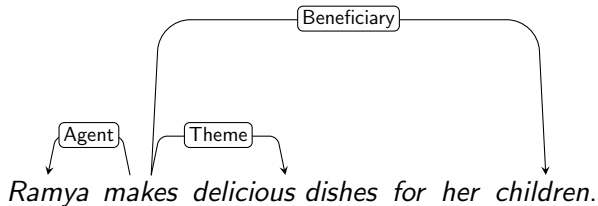


Figure: 4

Sense Relations

- **Synonym:** a word having the same or nearly the same meaning as another word.
e.g. choice — selection, complex — complicated
- **Antonym:** a word having the opposite meaning as another word.
e.g. happy vs. unhappy, heavy vs. light, long vs. short
- **Subordinate word (hyponym):** a word whose meaning is included in the meaning of another word.
red < color (red is subordinate to color), pony < horse < animal
- **Superordinate word (hypernym):** a word whose meaning includes meaning of another word. If X is superordinate to Y then Y is subordinate to X.
e.g. color > red, black, green, . . . ; animal > horse, tiger, . . .

- **Meronymy:** denotes part/member of something.
e.g. page is a meronym of book, wheel, engine, . are meronyms of car
- **Holonymy:** opposite of meronym
- **Homonymy:** a word having a different meaning, but the same form as another word.
e.g. bear (verb) vs. bear (noun, animal), bank (institution) vs. bank (river bank)

Homonymy and Synonymy – complementary notions

- synonyms – the same meaning, different forms.
- homonyms – the same form, different meanings.

Sentence Relations

Sentences can also be semantically related to one-another in a few different ways

- **Paraphrase:** They have the same truth conditions; if one is true, the other must also be true.
E.g. 'The boys like the girls' and 'The girls are liked by the boys'.
- **Mutual entailment:** Each sentence must be true for the other to be true.
E.g. 'John is married to Rachel' and 'Rachel is John's wife'
- **Asymmetrical entailment:** Only one of the sentences must be true for the other to be true, but that sentence may be true without the other sentence necessarily having to be true.
E.g. 'Rachel is John's wife' entails 'John is married' (but John is married does not entail Rachel being his wife)
- **Contradiction:** Sentences contradict each other when one sentence is true and the other cannot be true.
E.g. 'Rachel is an only child' and 'Rachel's brother is called Phil'.

Pragmatics and Discourse

Pragmatics and Discourse

Pragmatics

Pragmatics is about the use of language in context, where context includes both the linguistic and/or situational context of an utterance / text sentence.

Discourse

Discourse involves studying language beyond individual sentences.

Discourse Processing

Discourse refers to language beyond individual sentences.

- Discourse studies sequences of sentences that form a coherent conversation or text.
- It is crucial for understanding natural language in a broader context (paragraphs, conversations).
- Discourse analysis helps in understanding relationships between sentences.
- It involves understanding language at a higher level than just sentence structure.

Cohesion and Coherence in Discourse

- **Coherence** refers to the overall clarity and logical flow of ideas in a text or speech. It is how well ideas connect at a deeper level, making the text meaningful and understandable as a whole.
- **Cohesion** refers to the use of linguistic devices (such as conjunctions, pronouns, or repetition) that link sentences and paragraphs together at the surface level.
- **Coherence Example:**
 - ▶ *The weather is nice today. I will go for a walk.* (The ideas are logically connected and make sense together, showing a coherent flow.)
- **Cohesion Example:**
 - ▶ *John loves pizza. He eats it every day.* (The pronoun "He" and the noun "it" link the sentences cohesively.)
- Both coherence and cohesion are essential for effective communication in writing and speech.

Devices of Coherence and Cohesion

- Anaphora** : refers back to another word or phrase earlier in the text.
 It often involves pronouns.
John forgot his umbrella. **He** left it in the car.
- Connectives** : words or phrases that link sentences, clauses, or phrases together
 She wanted to go for a walk, **but** it started to rain.
- Ellipsis** : words are omitted when the phrase has to be repeated.
 A: Where are you going?
 B: To cinema
- Substitution** : word is not omitted, as in ellipsis, but is substituted for another, more general word.
 A: Which **ice-cream** would you like?
 B: I would like the pink **one**

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Yule, G. - *The Study of Language*, Cambridge

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Chomsky, N. - *Syntactic Structures* (1957)

Questions?



Thank
you